



NATURAL FIBER COMPOSITE · AHIMSA DESIGN

Only when we make the right
choices **do we achieve the right
results.**

TERMITE PROOF

WATER PROOF

FLAME RETARDANT

EASILY MACHINABLE

THERMOFORMABLE

ECO-FRIENDLY & RECYCLABLE

TECHNICAL GUIDELINES · REVISION 2026

Working with Indowud nfc.

Framing, fastening, edge protection, thermoforming and gluing — everything a workshop or site team needs to install nfc panels correctly, indoors and out.

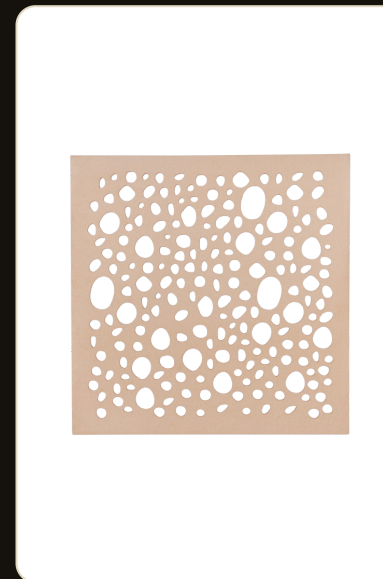
FRAMING & SUPPORT

FASTENING

EDGE SEALING

THERMOFORMING

GLUING



STEP 1 · BEFORE YOU START

A few principles that apply everywhere.

Indowud nfc machines like premium hardwood — but it is a homogeneous composite. Respect breathing gaps, seal cut edges, and support spans, and every install will last.

APPLIES TO All nfc products



Acclimatise & plan

Store flat on a level surface. Plan framing and fixings before cutting — nfc rewards a measured approach.



Always seal cut edges

Close micro-pores on every cut edge and the face with sealant, primer, laminate or edge-banding before exposure to water.



Leave breathing gaps

12 mm to walls, 5 mm at ceiling/floor, 3 mm between boards/trims, 4 mm between deck planks — for thermal movement.

Golden rule: never pour water on a raw board before its surface and edge micro-pores are closed. Do not use NC putty — use epoxy / solvent putty, primer or sealant.

STEP 2 · FRAMING & SUPPORT

Maximum support distances.

Indowud nfc is homogeneous in structure. For larger areas, provide constructive support / framing per the maximum support distances below (all values in mm). Higher-density grades span further.

APPLIES TO Boards (Neo / Create / Build) · decking · jaali · textured panels

BOARD THICKNESS	NEO · 625+ KG/CBM	CREATE · 725+ KG/CBM	BUILD · 825+ KG/CBM
6 mm	150	200	250
8 mm	200	250	350
12 mm	250	300	450
15/16 mm	300	350	550
18 mm	350	400	600
20 mm	400	450	650
25 mm	500	550	750

Ceilings: use 12 mm+ board on a grid support of 300 mm (Create/Build) or 200 mm (Neo).

Decking: use 25 mm+ on 300 mm support; leave a 4 mm gap between deck planks.

Suggested distances may increase or decrease depending on application and load.

STEP 3 · SHUTTERS, DOORS & CABINETS

Keep panels flat and balanced.

Bowing comes from unbalanced finishing and unsealed pores. Seal first, balance both faces, and support the spans.

APPLIES TO nfc board · nfc door · nfc frame

- 01 Fix a magnetic ball catch at the top and bottom of wardrobe / cabinet shutters. A straightener / stiffener is recommended.
- 02 Use box-type hinges at a span of every 300 mm.
- 03 Fix shutters only after overlaying laminate, veneer, edge-band / lipping — or applying epoxy / solvent putty, primer or sealant all around to close micro-pores and avoid bowing.
- 04 Ensure framing on all four sides with stiffeners for sliding doors / shutters.
- 05 For tall shutters, balance the finish on both faces to keep the panel flat.



STEP 4 · FASTENING

Excellent screw & nail holding.

High natural-fibre content gives nfc holding power that ordinary panels can't match. Choose the right fastener and it stays put.

APPLIES TO All boards · decking · fence · trims · frames



Choose the right screw

Use countersunk, wide-head, fully-threaded screws. Pilot-drill near edges. Gluing the edge before screwing adds joinery strength.



Outdoor fixings

Use stainless-steel screws for all exterior applications to avoid corrosion staining. Keep fixings back from the edge.



Nailing

nfc accepts nails like timber. For structural joints prefer screws + adhesive; reserve nails for trims and light fixing.



Conventional tools

Cut, drill, rout and sand with standard woodworking tools and carbide tips — no special equipment required.

STEP 5 · EDGE PROTECTION & SEALING

Close the pores, lock out water.

nfc is waterproof through its body, but freshly cut edges expose micro-pores. Sealing them is the single most important finishing step — especially outdoors.

APPLIES TO Every cut product — especially exterior: decking · fence · jaali

METHOD	BEST FOR
Edge-banding / lipping	Visible furniture & shutter edges
Laminate / veneer wrap	Decorative interior panels
Epoxy / solvent putty	Filling, levelling, sealing pre-paint
Primer + sealant	Paint-grade and exterior surfaces

Do not use NC putty. **Do not** expose raw, unsealed edges to standing water. Seal all four edges and both faces before exterior installation.



STEP 6 · THERMOFORMING

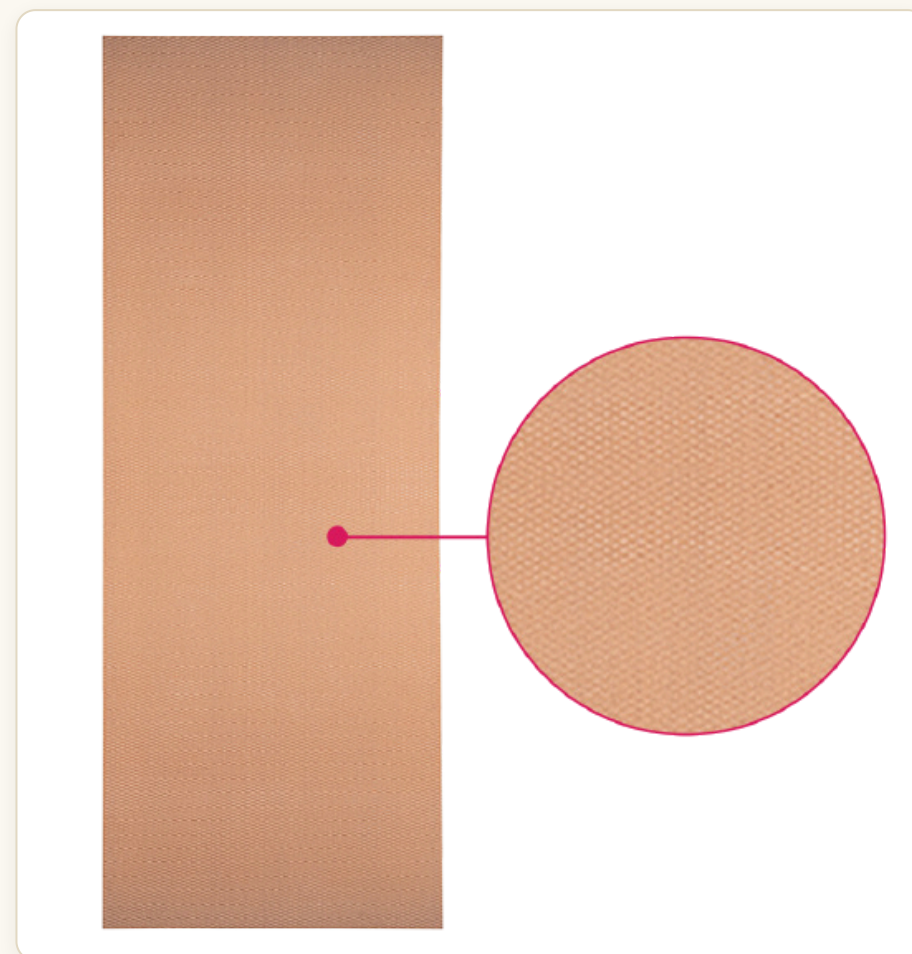
Shape it without compromise.

Thermoform nfc with a thermoforming machine, heat oven, heat gun or infrared heater. High natural-fibre content keeps density and dimensional stability uncompromised after forming.

APPLIES TO nfc board (Neo / Create) · textured panels

PARAMETER	VALUE
Heating temperature	140 – 160 °C
Heating time	1–2 min per mm of thickness (e.g. 18–36 min for 18 mm)
Locking time	5–20 min, by thickness & conditions
Cooling time	1–2 min per mm of thickness

Note: the panel may shrink as it is shaped. Do all cutting and finishing *after* the board takes its final form. Prepare a mould for even, repeatable shaping.



STEP 7 · PRINTING & FINISHING

Finish it any way you like.

nfc takes print, paint, laminate and veneer as readily as fine wood.

APPLIES TO Boards · doors · textured panels · trims



Printing

Any pattern prints via flatbed digital printer.
Ensure the board is dust-free first. Protect the print with a transparent sealant or varnish against abrasion.



Paint & polish

Paint, polish, stain or varnish directly once pores are sealed and primed. Balance the finish on both faces of unsupported panels.



Laminate & veneer

Overlay laminate or veneer with PUR or recommended adhesives. Seal the opposite face to keep the panel flat.

STEP 8 · GLUING

A strong, water-resistant bond.

We strongly recommend **NFC-GLU** with Indowud nfc — it bonds almost all surfaces, rough or smooth, porous or non-porous, faster.

APPLIES TO All products · NFC-GLU

RECOMMENDED ADHESIVES

NFC-GLU

PUR

HEATX

WP1

PROBOND

PLASTILOK

RELAM

Drying / locking time varies with temperature, humidity and surface. Clamp or weight the joint until set. Glued edges also improve screw-joinery strength.

Tip: for structural joints, combine adhesive with countersunk screws. Seal the glued edge afterward for exterior use.



QUICK REFERENCE

Keep this card on site.

Breathing gaps

12 mm to walls · 5 mm ceiling/floor · 3 mm between boards · 4 mm between deck planks

Ceilings

12 mm+ board · 300 mm grid (Create/Build) · 200 mm (Neo)

Edge sealing

Always seal cut edges before water exposure; never use NC putty

Hinges

Box-type every 300 mm; ball catch top & bottom of shutters

Decking

25 mm+ board · 300 mm support · 4 mm plank gap

Thermoforming

140–160 °C · cut & finish after final shape

TECHNICAL SUPPORT

info@indowud.com · www.indowud.com

Always test on a sample before full production.

TOGETHER

Let's build a world where beauty and responsibility go hand in hand.

TECHNICAL SUPPORT

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